

Masses and Planck Scale without ν

One chain, one anchor

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Masses and Planck Scale without ν

Abstract

The lepton ladder of the corpus writes the masses as $m_i = r_i \xi^{p_i} \nu$ and uses the electroweak scale ν as an intermediate quantity; the Planck scale is reached separately through the formula for G with the conversion factor C_{conv} . Read this way, the calculation seems to need two anchors, one for ν and one for E_P . It does not. Doc. 149 links ν and E_P directly: $\nu = E_P / (f^4 \cdot (\pi/2) \cdot 10)$, i.e. $\nu/E_P = \xi^4 / (5\pi) = 2.012 \times 10^{-17}$, -0.23% against the measured value **[K]**. Inserted into the ladder, ν drops out: $m_i = \frac{r_i}{5\pi} \xi^{p_i+4} E_P$ **[B]**. All three lepton masses, E_P , G , ℓ_P and L_0 then hang on a single chain, and one measured value as anchor fixes all numbers. With m_e as anchor one obtains $E_P = 1.237 \times 10^{19}$ GeV ($+1.34\%$), $G = 6.50 \times 10^{-11}$ (-2.6%), $\ell_P = 1.595 \times 10^{-35}$ m and $L_0 = 2.127 \times 10^{-39}$ m **[K]**. The deviations are the known bare residues of the ladder, not a second anchor.

On notation: here f is the base winding number $f = 1/\xi = 7500$, not a frequency. In natural units f is only another reading of the duality $T \cdot m = 1$: $f = t_P/t_0 = \Lambda_{T0}/E_P$ with $t_0 = \xi t_P$ and $\Lambda_{T0} = E_P/\xi$, and $\xi E_0^2 = 1$ gives $f = E_0^2$. As a dimensionless ratio, f cannot be set to 1 with justification (Doc. A135).

Status markers

- [B]** Proven algebraically (rearrangement without additional assumption).
- [K]** Confirmed numerically (check script).
- [S]** Speculative: a hint or conjecture, not yet derived or checked in the corpus.

Measured values (CODATA 2022, PDG) serve as anchor or as comparison values. $\hbar$, c and e are exact by definition in the SI; they fix only the units and are not an additional anchor.

1 Starting point

Two routes stand side by side in the corpus:

1. **Lepton ladder** (Docs. 006, 352): $m_i = r_i \xi^{p_i} v$ with $r = (4/3, 16/5, 25/9)$ and $p = (3/2, 1, 2/3)$ for e, μ, τ . The mass ratios $m_\mu/m_e = \frac{12}{5} \xi^{-1/2} = 207.85$ and $m_\tau/m_\mu = \frac{125}{144} \xi^{-1/3} = 16.99$ do not contain v .

2. **Gravitation** (Docs. 012, 013, 180): $G = \xi^2/(4m_e) \cdot C_{\text{conv}} \cdot K_{\text{frak}}$, and from it E_P and ℓ_P .

Taking v from m_e and E_P from the G formula uses two measured values: m_e and, through C_{conv} , the measured value of G . This is how Doc. 375 calculates. A chain meant to derive all quantities from ξ alone, however, may have only one anchor; all other numbers must follow. Doc. 149 supplies the link.

2 The relation between v and E_P

Doc. 149 derives the electroweak scale from the Planck energy: dilution over the f^4 cells of the four-dimensional lattice, projection onto the three-dimensional half-space ($\pi/2$), and a factor $1/10$ for the step to the electroweak scale:

$$v = \frac{E_P}{f^4 \cdot (\pi/2) \cdot 10} \quad \Longleftrightarrow \quad \frac{v}{E_P} = \frac{\xi^4}{5\pi} \quad \textbf{[B]} \quad (1)$$

with $f = 1/\xi$. The rearrangement is exact. The factor 10 is neither derived nor motivated but fitted to the measured value: in the working state of February 2026, $m_P/(f^4 \cdot \pi/2)$ gave 2467 GeV instead of 246 GeV, and the gap was closed with $1/10$ (Doc. 149). If the bare $v = 248.93$ GeV of the ladder is inserted instead of the measured v , the formula requires $10 K_{\text{frak}} \approx \pi^2$ instead of 10; the factor 10 would then be a mix-up of the bare ladder value with the SI comparison value. This is a choice and open **[S]**; Doc. 387 compares the variants.

Numerical value. $v/E_P = \xi^4/(5\pi) = 2.0120 \times 10^{-17}$. The measured value is $v/E_P = 246.22 \text{ GeV} / 1.22089 \times 10^{19} \text{ GeV} = 2.0167 \times 10^{-17}$; the deviation is -0.23% **[K]**. Since v/E_P is dimensionless, this number needs no anchor.

On the value in Doc. 149. There the calculation uses $f = 7491.91$ and gives $v = 246.71$ GeV. With $f = 1/\xi = 7500$ the same formula gives $v = 245.65$ GeV **[K]**. Both values are calculated correctly; they differ in the f inserted.

3 The masses without ν

Inserting $\nu = \xi^4 E_P / (5\pi)$ into the ladder gives

$$m_i = \frac{r_i}{5\pi} \xi^{p_i+4} E_P \quad [\mathbf{B}] \quad (2)$$

With the measured value $E_P = 1.22089 \times 10^{22}$ MeV for comparison:

Lepton	Formula	Coefficient	Value	vs. measurement
e	$\frac{4}{15\pi} \xi^{11/2} E_P$	0.08488	0.5043 MeV	−1.32 %
μ	$\frac{16}{25\pi} \xi^5 E_P$	0.20372	104.81 MeV	−0.80 %
τ	$\frac{5}{9\pi} \xi^{14/3} E_P$	0.17684	1780.9 MeV	+0.23 %

[K] The mass ratios are the same as in the ladder, since E_P and 5π cancel. The deviations combine the bare residue of the ladder against ν (Doc. 352) and the −0.23 % of the relation ν/E_P .

4 One chain, one anchor

The chain reads:

$$\xi \longrightarrow \frac{m_i}{E_P} = \frac{r_i}{5\pi} \xi^{p_i+4} \longrightarrow E_P \longrightarrow G = \frac{\hbar c^5}{E_P^2}, \quad \ell_P = \frac{\hbar c}{E_P}, \quad L_0 = \xi \ell_P. \quad (3)$$

Apart from the units, everything is fixed by ξ . A single measured value fixes the numbers. With $m_e = 0.51099895$ MeV as anchor:

Quantity	from anchor m_e	vs. measurement
E_P	1.2372×10^{19} GeV	+1.34 %
G	6.4995×10^{-11} m ³ kg ^{−1} s ^{−2}	−2.6 %
ℓ_P	1.5950×10^{-35} m	−1.32 %
$L_0 = \xi \ell_P$	2.1266×10^{-39} m	−1.32 %

[K] Since $G \propto E_P^{-2}$, the relative deviation of G is twice that of E_P .

Choice of anchor. The chain can start at any point; which quantity serves as anchor is a question of accuracy, not of derivation. The deviations of the other quantities then depend on the bare residue of the chosen entry point:

Anchor	E_P	G	ℓ_P, L_0
m_e	+1.34 %	−2.62 %	−1.32 %
m_μ	+0.81 %	−1.60 %	−0.80 %
m_τ	−0.23 %	+0.45 %	+0.23 %
ν	+0.23 %	−0.46 %	−0.23 %

[K] In every case exactly one measured value is used. The spread between the rows is the bare residue of the ladder; it shrinks to the extent that Doc. 352 resolves this residue.

5 Relation to the G formula with C_{conv}

The formula $G = \xi^2/(4m_e) \cdot C_{\text{conv}} \cdot K_{\text{frak}}$ from Doc. 012 gives 6.6745×10^{-11} , i.e. $+0.003\%$. This does not contradict the -2.6% above: $C_{\text{conv}} = 7.783 \times 10^{-3}$ is chosen so that the SI conversion matches the measured value of G . In this form m_e and G both enter as anchors. The one-anchor chain with m_e corresponds to $C_{\text{conv}} = 7.58 \times 10^{-3}$ **[K]**; the difference of 2.6% is exactly the residue of the chain. C_{conv} remains the SI conversion of G ; the chain only shows that it is not needed as a second measured value if the residue is accepted.

Doc. 375 calculates ν/E_P along this route (ν from m_e , E_P via C_{conv}) and obtains 2.0389×10^{-17} . The direct value $\xi^4/(5\pi) = 2.0120 \times 10^{-17}$ lies lower by the factor 1.0134; this is exactly the residue -1.32% of the electron mass in the chain **[K]**. Both routes are therefore consistent.

6 References in other documents

- **Doc. 149:** At the formula for ν it states that it gives 245.65 GeV with $f = 7500$ and means $\nu/E_P = \xi^4/(5\pi)$.
- **Doc. 180:** At ℓ_P it states that the value 1.616×10^{-35} m is the measured value used as anchor and that the one-anchor chain gives 1.595×10^{-35} m.
- **Doc. 375:** It states that ν/E_P follows directly as $\xi^4/(5\pi)$, without C_{conv} and without an anchor.

7 Result

In the mass calculation ν is an intermediate quantity and can be eliminated: $m_i = \frac{f_i}{5\pi} \xi^{p_i+4} E_P$ **[B]**. The lepton masses, E_P , G , ℓ_P and L_0 thus hang on one chain that is fixed by ξ apart from the units. A single measured value suffices as anchor; which one is free. The deviations of 0.2 to 1.3 % in E_P are the bare residues of the ladder and of the relation ν/E_P (-0.23%), not the consequence of a missing anchor. The factor 10 in this relation is the open point of the chain **[S]**.

Check script: 2/python/Dok383_Skripte/pruef_383_ein_anker.py, 20/20.

Bibliography

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- [2] Particle Data Group (S. Navas et al.), *Review of Particle Physics*, Phys. Rev. D 110 (2024) 030001.